

ISRA CHAUDRY

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OBJECTIVE

Third year BS Cybersecurity student at GIKI with hands on experience in cryptographic systems engineering, threat modelling, network security, and blockchain-based identity platforms. Experienced in deploying secure systems in real-world environments including healthcare infrastructure. Seeking a paid internship in Cybersecurity, Security Engineering, or SOC/Defensive Security.

EDUCATION

Ghulam Ishaq Khan Institute (GIKI) BS Cybersecurity, Final Year

2023 – 2027

WORK EXPERIENCE

Networking Intern at Ayub Medical Hospital, Abbottabad

Summer 2025

- Configured and secured hospital network infrastructure, implemented firewall rules, access controls, and network segmentation to protect sensitive patient health records across critical medical systems.
- Applied real-world network defense practices in a live healthcare environment serving 1,500+ beds, gaining hands-on exposure to securing critical public infrastructure.

CYBERSECURITY & AI PROJECTS

SecureVerse, Decentralized Identity & Secure Communication Platform

- Engineered a full-stack zero-trust identity platform with 3-factor authentication: blockchain DIDs anchored on Ethereum Sepolia, TOTP 2FA via pyotp, and RS256 JWT tokens using authlib, replacing python jose mid-project after discovering two active CVEs (2024).
- Implemented end-to-end encrypted messaging, HMAC-SHA256 tamper-evident audit log chain anchored on-chain, and RBAC enforced both on-chain (Solidity) and in backend middleware.
- Conducted full security testing lifecycle: threat modelling with IRIS and STRIDE, static analysis with Bandit and Pylint, and dynamic testing with Postman, verified resistance to 20 attack vectors including JWT confusion, SQL injection, brute force, and token replay.

High-Entropy Cryptographic Key Generation Using Lava Lamp

- Built a hardware-based true random number generator using live lava lamp video frames captured via OpenCV, extracting per-pixel RGB variance from unpredictable fluid motion as a physical entropy source.
- Processed entropy in C++ using SHA-256 hashing and XOR mixing to derive cryptographic keys with measurably higher unpredictability than standard software-based PRNG methods.

AI-Based Phishing Email Detection System

- Fine-tuned DistilBERT transformers on a phishing dataset achieving 94% precision and 91% recall on a 5,000-email test set, applied to detect social engineering attacks targeting non-technical users.
- Deployed as a live Streamlit web application with dual inference modes for real-time threat classification.

LoRaWAN Network Security Analysis (Ongoing Research)

- Identifying and documenting LoRaWAN protocol vulnerabilities including replay attacks, join procedure weaknesses, and insufficient end-to-end encryption across IoT deployments.
- Proposing concrete protocol enhancements to key management and encryption mechanisms for resource-constrained IoT networks used in public infrastructure.

TECHNICAL SKILLS

Security: Cryptography (AES-256, Blowfish, 3DES, IDEA, NaCl), Network Security, Firewall Config, Penetration Testing, Threat Modelling (STRIDE, IRIS), Static Analysis (Bandit, Pylint), LoRaWAN, Digital Forensics

Blockchain: Solidity, Ethereum, Smart Contracts, Blockchain DIDs, Web3.py, JWT (RS256), TOTP 2FA

Languages: Python, C++, SQL, JavaScript, OOP, Data Structures & Algorithms

Tools: Postman, Git, Linux, FastAPI, React, OpenCV, PostgreSQL, SQLite

AI / ML: DistilBERT / HuggingFace, TensorFlow, PyTorch, NLP, LangGraph, LangChain

ACHIEVEMENTS & LEADERSHIP

- Vice President, Netronix (Networking & Gaming Society, GIKI), managing internet and Wi-Fi infrastructure across 13 student hostels, troubleshooting network issues, and organizing society events.